AP CALCULUS AB

Unit 1 • Practice • Answer key

Limits and Continuity

AP-style practice • Multiple choice and free response • Answer key with explanations

About this practice set

The questions follow the format of the AP Calculus AB exam. Part A and Part B are multiple choice; Part C is free response. Each question is labeled **NO CALCULATOR** or **CALCULATOR**. Suggested timing: Part A 40 min • Part B 8 min • Part C 45 min.

Part A • Multiple choice • No calculator

Directions • Part A

A calculator may **not** be used on this part. Solve each problem, then choose the best of the four answer choices. Unless otherwise stated, the domain of a function f is the set of all real numbers x for which $f(x)$ is a real number. Some questions refer to a graph: read the values you need from the grid.

Question 1 (1.6)

NO CALCULATOR

$\lim_{x\to 3} \dfrac{x^2-9}{x^2-2x-3}$ is

(A) 0

(B) $\dfrac{3}{2}$

(C) 3

(D) nonexistent

Solution

Answer: (B). $\dfrac{(x-3)(x+3)}{(x-3)(x+1)} = \dfrac{x+3}{x+1}$ for $x \neq 3$, and $\dfrac{6}{4} = \dfrac{3}{2}$.

Question 2 (1.6)

NO CALCULATOR

$\lim_{x\to 0} \dfrac{\sqrt{x+9}-3}{x}$ is

(A) 0

(B) $\dfrac{1}{3}$

(C) $\dfrac{1}{6}$

(D) 3

Solution

Answer: (C). Multiply by the conjugate: $\dfrac{(x+9)-9}{x\left(\sqrt{x+9}+3\right)} = \dfrac{1}{\sqrt{x+9}+3} \to \dfrac{1}{6}$.

Question 3 (1.8)

NO CALCULATOR

$\lim_{x\to 0} \dfrac{\sin(5x)}{x}$ is

(A) 5
(B) 1
(C) $\frac{1}{5}$
(D) 0

Solution
Answer: (A). $\frac{\sin 5x}{x} = 5 \cdot \frac{\sin 5x}{5x} \to 5 \cdot 1 = 5.$

Question 4 (1.15) NO CALCULATOR

$\lim_{x\to\infty} \frac{4x^3 - x}{2x^3 + 7x^2}$ is

(A) 0
(B) $\frac{4}{7}$
(C) ∞
(D) 2

Solution
Answer: (D). Equal degrees: the limit is the ratio of the leading coefficients, $\frac{4}{2} = 2$.

Question 5 (1.15) NO CALCULATOR

$\lim_{x\to-\infty} \frac{2x}{\sqrt{x^2+5}}$ is

(A) 2
(B) -2
(C) 0
(D) $-\infty$

Solution
Answer: (B). For $x < 0$, $\sqrt{x^2+5} \approx |x| = -x$, so the quotient approaches $\frac{2x}{-x} = -2$.

Question 6 (1.7) NO CALCULATOR

$\lim_{x\to 2^-} \frac{|x-2|}{x-2}$ is

(A) 1
(B) 0
(C) -1
(D) nonexistent

Solution
Answer: (C). For $x < 2$, $|x-2| = -(x-2)$, so the quotient equals -1.

Question 7 (1.4) NO CALCULATOR

x	0.9	0.99	0.999	1.001	1.01	1.1
$f(x)$	2.71	2.97	2.997	3.003	3.03	3.31

Based on the table, which is the best estimate of $\lim_{x \to 1} f(x)$?

(A) 2.997
(B) 3
(C) 3.003
(D) The limit does not exist.

Solution
Answer: (B). The values approach 3 from both the left and the right.

Question 8 (1.11)

NO CALCULATOR

Let $f(x) = \begin{cases} x^2 + 1, & x < 1 \\ 3x - 1, & x \geq 1. \end{cases}$ Which statement is true?

(A) $\lim_{x \to 1} f(x)$ does not exist.
(B) $\lim_{x \to 1} f(x)$ exists, but f is not continuous at $x = 1$.
(C) f is continuous at $x = 1$.
(D) $f(1)$ is undefined.

Solution
Answer: (C). $\lim_{x \to 1^-} f(x) = 2$, $\lim_{x \to 1^+} f(x) = 2$ and $f(1) = 3 - 1 = 2$. All three conditions of continuity hold.

Question 9 (1.13)

NO CALCULATOR

For what value of k is $f(x) = \begin{cases} kx + 3, & x \leq 2 \\ x^2 - k, & x > 2 \end{cases}$ continuous at $x = 2$?

(A) $-\frac{1}{3}$
(B) $\frac{1}{3}$
(C) 1
(D) 7

Solution
Answer: (B). Left: $2k + 3$. Right: $4 - k$. Then $2k + 3 = 4 - k$, so $3k = 1$ and $k = \frac{1}{3}$.

Question 10 (1.10)

NO CALCULATOR

What type of discontinuity does $g(x) = \dfrac{x^2 - 4}{x - 2}$ have at $x = 2$?

(A) Jump
(B) Infinite
(C) Removable
(D) None; g is continuous at $x = 2$.

Solution
Answer: (C). $g(2)$ is undefined, but $\lim_{x \to 2} g(x) = \lim_{x \to 2}(x + 2) = 4$ exists: a removable discontinuity.

Question 11 (1.14) NO CALCULATOR

Which line or lines are vertical asymptotes of the graph of $h(x) = \dfrac{x+1}{x^2-1}$?

(A) $x = 1$ only
(B) $x = -1$ only
(C) $x = 1$ and $x = -1$
(D) $y = 0$ only

Solution
Answer: (A). $h(x) = \dfrac{x+1}{(x-1)(x+1)} = \dfrac{1}{x-1}$ for $x \neq -1$. At $x = -1$ there is a hole (the limit is $-\frac{1}{2}$); at $x = 1$ the one-sided limits are infinite. $y = 0$ is a horizontal asymptote.

Question 12 (1.8) NO CALCULATOR

$\lim\limits_{x\to 0} x^2 \cos\left(\dfrac{1}{x}\right)$ is

(A) 1
(B) 0
(C) -1
(D) nonexistent

Solution
Answer: (B). $-x^2 \leq x^2 \cos\frac{1}{x} \leq x^2$ and both bounds approach 0. By the squeeze theorem, the limit is 0.

Question 13 (1.16) NO CALCULATOR

The function f is continuous on $[1,5]$ with $f(1) = -2$ and $f(5) = 6$. Which statement must be true?

(A) $f(c) = 8$ for some c in $(1,5)$.
(B) $f(3) = 2$.
(C) f is increasing on $[1,5]$.
(D) $f(c) = 4$ for some c in $(1,5)$.

Solution
Answer: (D). 4 is between -2 and 6, and f is continuous, so the IVT guarantees such a c. Nothing forces $f(3) = 2$ or monotonicity, and 8 is not between the endpoint values.

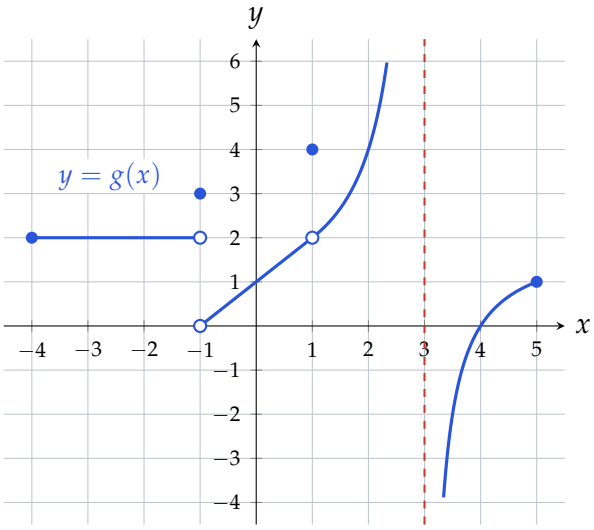
Question 14 (1.15) NO CALCULATOR

$\lim\limits_{x\to\infty} \dfrac{3e^x + x^2}{e^x - 1}$ is

(A) 0
(B) 1
(C) 3
(D) ∞

Solution
Answer: (C). e^x dominates x^2 and constants as $x \to \infty$. Dividing by e^x: $\dfrac{3 + x^2 e^{-x}}{1 - e^{-x}} \to \dfrac{3+0}{1-0} = 3$.

Questions 15 to 19 refer to the graph of the function g below.



The domain of g is $[-4,3) \cup (3,5]$. The line $x = 3$ is a vertical asymptote.

Question 15 (1.3 • graph) NO CALCULATOR

Which of the following limits does **not** exist?

(A) $\lim_{x \to 1} g(x)$
(B) $\lim_{x \to -1^+} g(x)$
(C) $\lim_{x \to -1} g(x)$
(D) $\lim_{x \to 4} g(x)$

Solution
Answer: (C). At $x = -1$ the left-hand limit is 2 and the right-hand limit is 0. The other limits are 2, 0 and 0.

Question 16 (1.3 • graph) NO CALCULATOR

What is $\lim_{x \to 1} g(x)$?

(A) 4
(B) 2
(C) 1
(D) It does not exist.

Solution
Answer: (B). Both pieces approach the open circle at $(1,2)$. The filled dot $g(1) = 4$ does not affect the limit.

Question 17 (1.14 • graph) NO CALCULATOR

Which statement about the behavior of g near $x = 3$ is true?

(A) $\lim_{x \to 3^-} g(x) = +\infty$ and $\lim_{x \to 3^+} g(x) = -\infty$
(B) $\lim_{x \to 3^-} g(x) = -\infty$ and $\lim_{x \to 3^+} g(x) = +\infty$
(C) $\lim_{x \to 3} g(x) = +\infty$
(D) $\lim_{x \to 3} g(x) = 2$

Solution
Answer: (A). To the left of $x = 3$ the graph rises without bound; to the right it falls without bound.

Question 18 (1.10 • graph) NO CALCULATOR

At which value of x does g have a removable discontinuity?

(A) $x = -1$
(B) $x = 3$
(C) $x = -4$
(D) $x = 1$

Solution
Answer: (D). At $x = 1$, $\lim_{x\to 1} g(x) = 2$ exists but $g(1) = 4$. At $x = -1$ the discontinuity is a jump; at $x = 3$ it is infinite.

Question 19 (1.9 • graph) NO CALCULATOR

What is $\lim_{x\to 0} g\big(g(x)\big)$?

(A) 1
(B) 4
(C) 2
(D) It does not exist.

Solution
Answer: (C). As $x \to 0$, $g(x) = x + 1 \to 1$, taking values close to 1 but not equal to 1 (from below when $x < 0$, from above when $x > 0$). As the input approaches 1 from either side, g approaches 2. So the limit is 2, not $g(1) = 4$.

Part B • Multiple choice • Graphing calculator

Directions • Part B

A graphing calculator is **required** for the questions in this part. If the exact value of the correct answer is not one of the choices, choose the one that best approximates it.

Question 20 (1.4) CALCULATOR

Using a table of values or a graph, what is $\lim_{x\to 0} \dfrac{2^x - 1}{x}$, to three decimal places?

(A) 0.500
(B) 0.693
(C) 1.000
(D) 1.443

Solution
Answer: (B). Values at $x = \pm 0.001$ are about 0.6929 and 0.6934, so the limit is about 0.693 (it equals $\ln 2$).

Question 21 (1.16) CALCULATOR

How many values of c in the open interval $(0,2)$ satisfy $c^3 - 4c + 1 = 0$?

(A) None
(B) One
(C) Two
(D) Three

Solution
Answer: (C). Graphing $y = x^3 - 4x + 1$ shows zeros at $x \approx -2.115$, 0.254 and 1.861. Two of them are in $(0,2)$.

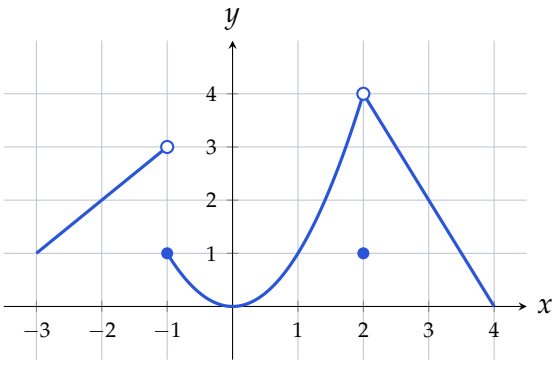
Part C • Free response

Directions • Part C

Show all of your work: your answers are graded on the correctness and completeness of your methods as well as on the final answers. Justify answers with mathematical reasons, naming the theorem or definition you use. Use a calculator only on the question labeled **CALCULATOR**. Decimal answers should be correct to three places after the decimal point.

Question 22 (Graph analysis) NO CALCULATOR

The graph of the function f on $[-3,4]$ is shown. It consists of a line segment for $-3 \le x < -1$, the parabola $y = x^2$ for $-1 \le x < 2$, and a line segment for $2 < x \le 4$. Also $f(-1) = 1$ and $f(2) = 1$.



(a) Find $\lim_{x \to -1^-} f(x)$ and $\lim_{x \to -1^+} f(x)$. Does $\lim_{x \to -1} f(x)$ exist? Give a reason for your answer.
(b) Find $\lim_{x \to 2} f(x)$. Is f continuous at $x = 2$? Name the type of discontinuity, if any, and explain how $f(2)$ could be redefined to make f continuous at $x = 2$.
(c) Find $\lim_{x \to 3} f(f(x))$. Show the work that leads to your answer.
(d) $f(1) = 1$ and $f(3) = 2$. Explain why the Intermediate Value Theorem cannot be used to guarantee a value c in $[1,3]$ with $f(c) = 1.5$. Does such a value exist? Justify.

Solution
(a) $\lim_{x \to -1^-} f(x) = 3$ and $\lim_{x \to -1^+} f(x) = (-1)^2 = 1$. The one-sided limits are not equal, so $\lim_{x \to -1} f(x)$ does not exist.
(b) $\lim_{x \to 2^-} f(x) = 2^2 = 4$ and $\lim_{x \to 2^+} f(x) = 8 - 2(2) = 4$, so $\lim_{x \to 2} f(x) = 4$. But $f(2) = 1 \neq 4$, so f is not continuous at $x = 2$. This is a removable discontinuity; redefining $f(2) = 4$ makes f continuous at $x = 2$.
(c) For x near 3, $f(x) = 8 - 2x \to 2$. As $x \to 3^-$, $f(x) \to 2$ from above; as $x \to 3^+$, $f(x) \to 2$ from below. In both cases $f(f(x))$ approaches $\lim_{y \to 2} f(y) = 4$ (from part (b)).

So $\lim_{x\to 3} f(f(x)) = 4$. (Note that this is not $f(2) = 1$.)
(d) The IVT requires f to be continuous on $[1,3]$, but f is not continuous at $x = 2$. Such a value does exist: on $[1,2)$, $f(x) = x^2 = 1.5$ when $c = \sqrt{1.5} \approx 1.22$, which is in $[1,3]$.

Question 23 (Algebraic limits and continuity) NO CALCULATOR

Let $g(x) = \begin{cases} \dfrac{x^2+x-6}{x-2}, & x < 2 \\ cx+1, & x \geq 2 \end{cases}$ where c is a constant, and let $h(x) = \dfrac{x^2+x-6}{x^2-4}$.

(a) Find $\lim_{x\to 2^-} g(x)$. Show the work that leads to your answer.
(b) Find the value of c for which g is continuous at $x = 2$. Justify your answer using the definition of continuity.
(c) Find all values of x where h is not continuous, and classify each discontinuity. Justify with limits.
(d) Find $\lim_{x\to\infty} h(x)$ and $\lim_{x\to -2^+} h(x)$.

Solution
(a) For $x < 2$: $\dfrac{(x+3)(x-2)}{x-2} = x+3$, so $\lim_{x\to 2^-} g(x) = 5$.
(b) $g(2) = 2c+1$ and $\lim_{x\to 2^+} g(x) = 2c+1$. For continuity we need $\lim_{x\to 2^-} g(x) = \lim_{x\to 2^+} g(x) = g(2)$: $2c+1 = 5$, so $c = 2$. Then $g(2) = 5$ and $\lim_{x\to 2} g(x) = 5 = g(2)$.
(c) $h(x) = \dfrac{(x+3)(x-2)}{(x+2)(x-2)}$ is not defined at $x = 2$ and $x = -2$. At $x = 2$: $\lim_{x\to 2} h(x) = \lim_{x\to 2} \dfrac{x+3}{x+2} = \dfrac{5}{4}$ exists: removable discontinuity. At $x = -2$: the numerator of $\dfrac{x+3}{x+2}$ approaches 1 and the denominator approaches 0, so $\lim_{x\to -2^-} h(x) = -\infty$ and $\lim_{x\to -2^+} h(x) = +\infty$: infinite discontinuity (vertical asymptote $x = -2$).
(d) $\lim_{x\to\infty} h(x) = 1$ (equal degrees, leading coefficients 1 and 1). $\lim_{x\to -2^+} h(x) = +\infty$.

Question 24 (Table and theorems) CALCULATOR

The temperature W, in degrees Celsius, of a greenhouse is a continuous function of time t, in hours, for $0 \leq t \leq 10$. Selected values are given in the table.

t (hours)	0	3	5	8	10
$W(t)$ (°C)	20	24	31	27	22

(a) Must there be a time t in $[0,10]$ at which $W(t) = 30$? Justify your answer.
(b) Explain why there must be at least two times in $[0,10]$ at which $W(t) = 25$.
(c) Use the data to estimate the rate at which the temperature is changing at $t = 4$ hours. Show the computation and give units.
(d) A function f satisfies $4x - x^2 \leq f(x) \leq x^2 - 4x + 8$ for all x. Find $\lim_{x\to 2} f(x)$, and name the theorem you use.

Solution
(a) Yes. W is continuous on $[3,5]$, $W(3) = 24$ and $W(5) = 31$. Since $24 < 30 < 31$, the IVT guarantees a t in $[3,5]$ with $W(t) = 30$.
(b) W is continuous on $[3,5]$ with $24 < 25 < 31$, so $W(t) = 25$ for some t in $[3,5]$. W is continuous on $[8,10]$ with $22 < 25 < 27$, so $W(t) = 25$ for some t in $[8,10]$. These intervals do not overlap, so there are at least two such times.
(c) $\dfrac{W(5) - W(3)}{5-3} = \dfrac{31-24}{2} = 3.5$ degrees Celsius per hour.
(d) At $x = 2$: $4(2) - 2^2 = 4$ and $2^2 - 4(2) + 8 = 4$. Both bounding functions are polynomials, so $\lim_{x\to 2}(4x - x^2) = 4$ and $\lim_{x\to 2}(x^2 - 4x + 8) = 4$. By the squeeze theorem, $\lim_{x\to 2} f(x) = 4$.